RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Team ID	**SS14 Neocognitron**			
Semester	**6^{th}**			
Course :	**Interdisciplinary Project**			
Team Details		**Program**	**USN**	**Name**
	1	**AIML**	**1RV23AI011**	**Ahibhruth A**
	2	**AIML**	**1RV24AI406**	**L Moryakantha**
	3	**ECE**	**1RV23EC196**	**Pavan S Aradhya**
	4	**ETE**	**1RV23ET050**	**Srivatsa B Janadri**
Project Title	Machine Learning Approach for Fog Detection and Visibility Assessment on Roads			
Center of Excellence				
Internal Guide				
Name Designation & Department	**Dr Keshav M** **Assistant Professor** **Department of Mechanical Engineering**			

INTRODUCTION:

Road accidents caused by low visibility, fog, and poor road infrastructure are a major safety concern on highways and urban roads. This system uses artificial intelligence, computer vision, and IoT-based sensors to monitor road conditions in real time. It detects hazards such as fog, potholes, and traffic signs, and predicts fog formation using environmental data. The system analyzes traffic conditions to generate dynamic risk assessments and safety alerts through a centralized dashboard. This approach helps improve driver awareness and supports traffic authorities in enhancing road safety.

OBJECTIVES:

1. **Predict fog formation using Machine Learning models trained on weather parameters.**
2. **Detect road hazards such as potholes and traffic signs using FFA-Net (Real Time Dehazing Model) + YOLOv8 (Deep Learning Object-Detection Model).**
3. **Provide real-time alerts and visualization through an interactive dashboard, enabling drivers and traffic authorities to make informed decisions and improve road safety.**

METHODOLOGY:

1. **Fog Prediction using Machine Learning**
   Environmental data such as temperature, humidity, light intensity, and visibility are collected using IoT sensors. This data is preprocessed and used to train machine learning models to predict fog formation in advance.

Mathematical Formula:

$$x = [T, H, L] \in \mathbb{R}^3$$

$$P(fog) = \sigma(w^T x + b)$$

$$fog\_density = \frac{1}{L + \epsilon}$$

Metrics: Accuracy, Precision, Recall, F1-score

2. **Image Enhancement for Low Visibility**
   Captured road images are processed using **FFA-Net (Feature Fusion Attention Network)** to remove haze and enhance visibility, improving the accuracy of subsequent detection models.

$$I_{clear} = FFA(I_{foggy})$$

$$L = ||I_{clear} - I_{groundtruth}||^2$$

   Metrics: PSNR, SSIM

3. **Road Hazard Detection using Deep Learning**
   Enhanced images are passed through **YOLOv8**, a real-time object detection model, to identify and classify road hazards such as potholes, traffic signs, and road humps.

$$Z = W * I + b$$

$$A = \max(0, Z)$$

$$IoU = \frac{Area(B \cap G)}{Area(B \cup G)}$$

$$Confidence = P(object) \times IoU$$

   Metrics :mAP,IOU,Precision , Recall

4. **Sensor–Vision Data Fusion**
   Outputs from LDR sensors and computer vision models are combined to create a comprehensive understanding of road conditions, enabling more accurate analysis of visibility and hazards.

$$h = [CNN(I), s], \quad s = [L]$$

$$R = w_1 \cdot CNN(I) + w_2 \cdot s$$

Metrics: Improvement in F1-score, fusion accuracy

5. **Risk Assessment and Decision Engine**
 A risk model integrates fog prediction, hazard detection, and traffic-related factors to compute a dynamic risk score and generate appropriate driving recommendations.

$$Risk = \alpha \cdot Fog + \beta \cdot Hazard + \gamma \cdot Visibility$$

$$Risk \in [0, 1]$$

Example:

$$Risk = 0.4(Fog) + 0.3(Pothole) + 0.3(LDR)$$

Metrics: Risk prediction accuracy, correlation score

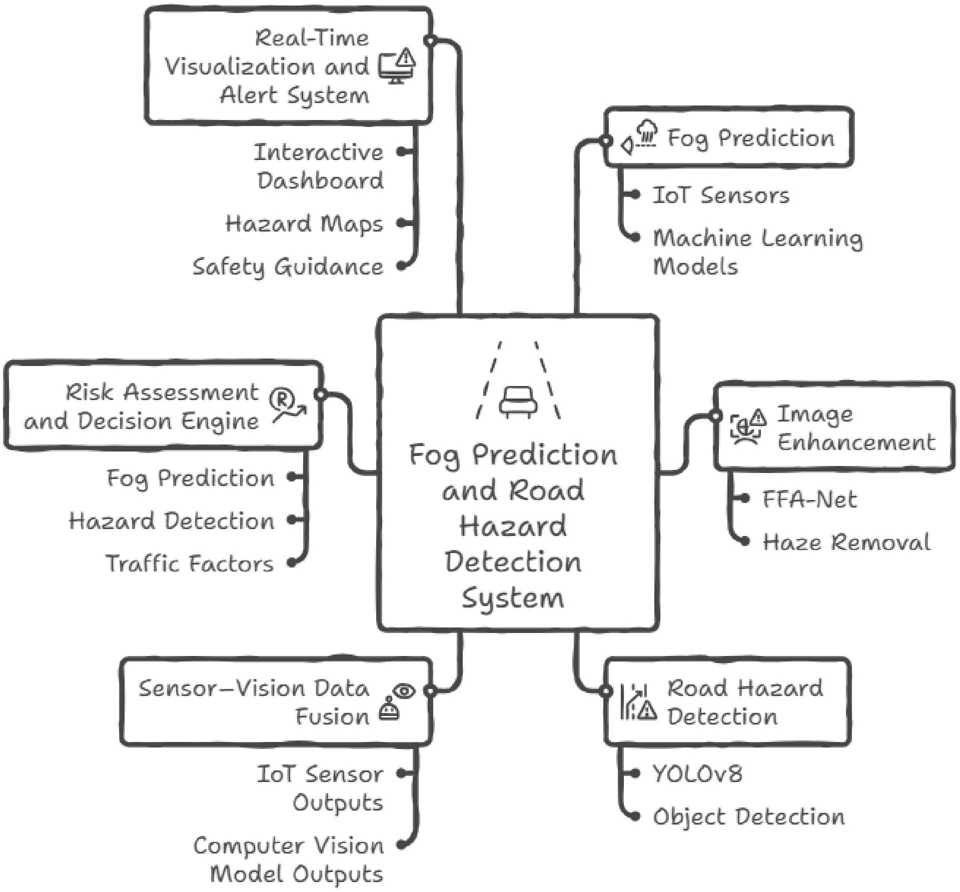
6. **Real-Time Visualization and Alert System**
 The processed data and predictions are displayed through an interactive dashboard, providing real-time alerts, hazard maps, and safety guidance for drivers and traffic authorities.

Mathematical formulation:

$$T_{total} = T_{capture} + T_{processing} + T_{alert}$$

$$FPS = \frac{Frames}{Second}$$

Metrics: Latency (<250 ms), FPS (15–20), alert accuracy



Software Requirements

Category	Tools / Technologies	Purpose
Programming Languages	**Python, JavaScript**	**Python for AI/ML and backend processing; JavaScript for frontend development**
Web Development Frameworks	**React / Next.js, Django / Flask**	**Building interactive dashboard and backend APIs**
AI / ML Frameworks	**TensorFlow, PyTorch, Scikit-learn**	**Training models for fog prediction, hazard detection, and risk analysis**

Computer Vision Libraries	**FFA-Net, OpenCV, YOLOv8**	**Image preprocessing, dehazing, and detection of potholes, fog, and traffic signs**
ML Algorithms	**XGBoost, CNN with Activation Functions of Leaky ReLU, Sigmoid, TanH**	**Fog prediction, time-series forecasting, risk modeling, and image-based detection**
Data Processing	**Pandas, NumPy**	**Data cleaning, normalization, and transformation**
Data Visualization	**Matplotlib, Plotly, Seaborn**	**Visualization of trends, predictions, and hazard insights**
Database	**PostgreSQL**	**Storing sensor data, predictions, hazard logs, and alerts**
Mapping Tools	**Google Maps API**	**Displaying road hazard maps and fog heatmaps**
Communication Protocols	**MQTT, WebSockets, REST APIs**	**Data transmission between ESP32 devices and cloud**
Annotation Tools	**LabelIing**	**Bounding box annotation for training datasets**

HARDWARE REQUIREMENTS:

Category	Hardware Components	Purpose
Edge Computing Device	**ESP32**	**Acts as a roadside processing unit to collect sensor data and perform edge-level computations**

Camera Module	**HD USB Camera**	**Captures road images or video for fog detection, pothole detection, and traffic sign recognition**
Light Sensor	**LDR (Light Dependent Resistor)**	**Measures ambient light intensity to estimate visibility conditions**
GPS Sensor	**NEO 6M**	**Tracks the Latitude and longitude of the system**

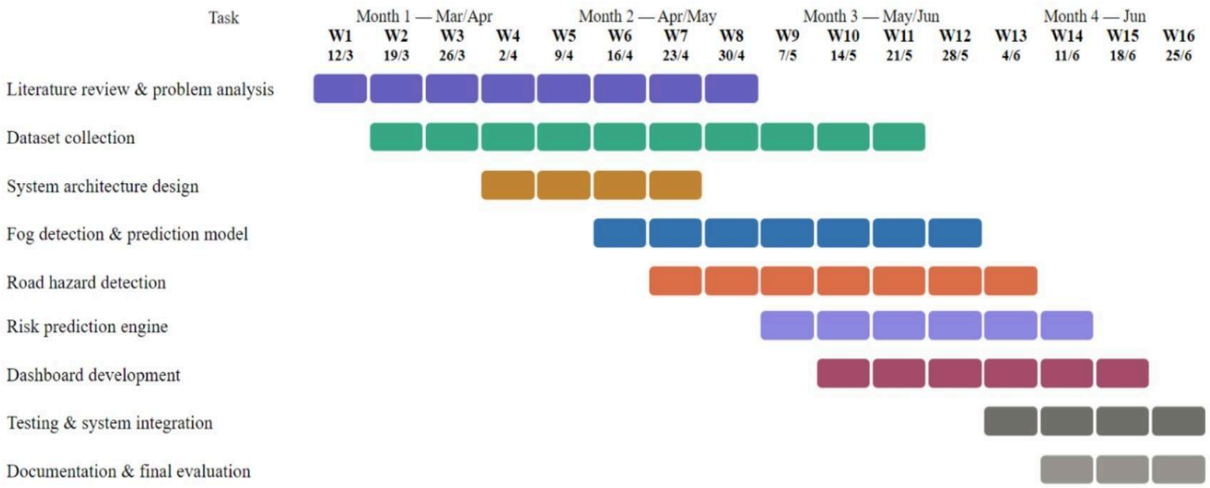
INTERDISCIPLINARY RELEVANCE:

The **AEGIS-RS** system demonstrates interdisciplinary collaboration between Artificial Intelligence & Machine Learning (AIML) and Electronics & Communication Engineering (EC) to enhance vehicle safety on road under low-visibility conditions. The EC component focuses on sensor-based data acquisition, embedded systems, and V2I2V communication system. The AIML component processes this data using machine learning and computer vision models to detect hazards, and analyze traffic conditions. By integrating real-time sensing with intelligent data analysis, the system provides proactive hazard monitoring and safety guidance for smarter transportation infrastructure.

INNOVATION / CONTRIBUTION TO THE FIELD:

The **AEGIS-RS** framework introduces an integrated approach that combines AI-driven computer vision, machine learning, and V2I2V communication for real-time road hazard monitoring and fog prediction. Unlike traditional reactive systems, it provides predictive risk analysis and intelligent safety guidance by analyzing environmental conditions, road hazards, and traffic data. This contributes to the development of smart transportation infrastructure and proactive accident prevention systems.

TIMELINE (GANTT CHART):



Signatures:

	Guide	HOD	Dean Academics
Name:	**Dr. Keshav M**	**Dr. Shanmukha Nagaraj**	**Dr. M V Renukadevi**
Designation:	**Assistant Professor**	**Professor and HoD**	
Department:	**ME**	**ME**	